



Spatio-Temporal Graph Neural Network for Motor Movement Classification Using 64-Channel EEG Signals

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Abstract

This study evaluates leakage-safe baselines and a hybrid spatio-temporal graph neural network (ST-GNN) for subject-independent classification of left- versus right-fist motor execution from 64-channel EEG. From the PhysioNet EEG Motor Movement/Imagery Dataset, 4,548 accepted trials from 108 subjects were divided by subject into training, validation, and test sets with no participant overlap. Classical models, XGBoost, an MLP, CSP-LDA, and a compact 1D-CNN were evaluated using accuracy, balanced accuracy, macro-F1, Cohen’s kappa, MCC, ROC-AUC, log loss, and execution time. Logistic regression was the predeclared classical baseline to beat, with validation macro-F1 of 0.6602. The best honest unseen-subject result was obtained by the 1D-CNN: 65.96% test accuracy and 0.6596 macro-F1. The proposed hybrid ST-GNN achieved 50.99% accuracy and 0.4975 macro-F1, indicating underfitting. The result shows that graph structure alone did not improve generalization under the tested settings and motivates graph, feature, and optimization ablations.

Keywords: EEG; motor execution; subject-independent classification; graph neural network; ST-GNN; 1D-CNN; leakage prevention.

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1. Introduction

Electroencephalography (EEG)-based brain-computer interfaces aim to infer a user’s intended or executed action from scalp-recorded neural activity. Left- and right-hand motor tasks are important because they support neurorehabilitation, assistive control, and the study of sensorimotor lateralization. The classification problem is difficult: EEG is noisy, non-stationary, spatially distributed, sensitive to artifacts, and highly variable between people. A model can therefore appear strong under a random trial split while failing when every test subject is unseen.

Most deep EEG classifiers learn temporal and channel patterns through convolution, recurrence, or attention. These methods are effective, but electrode relationships are often represented only implicitly. Graph neural networks (GNNs) offer a natural alternative because every electrode can be treated as a node and channel relationships can be represented as weighted edges. Nevertheless, a graph model is only scientifically useful when its adjacency is defensible, its preprocessing is leakage-safe, and its performance is compared against strong non-graph baselines under the same split.

The research gap identified from the prior review has four parts. First, many studies report results under incompatible validation protocols, making unseen-subject generalization difficult to assess. Second, graph-based EEG methods often rely on either fixed electrode geometry or data-derived connectivity, rather than combining both while estimating functional edges from training subjects only. Third, fewer studies fuse raw temporal EEG with interpretable node-level spectral and temporal features in one graph architecture. Fourth, negative or below-target GNN results are rarely discussed in enough detail to distinguish architecture failure, optimization failure, and genuine lack of benefit.

This report addresses that gap through a reproducible subject-wise experiment. It evaluates a broad baseline set, identifies the baseline to beat using validation data, defines a hybrid anatomical-functional graph, proposes a dual-branch ST-GNN, and reports the first results without altering labels or tuning on the test set.

2. Related Work

Five representative papers motivate the proposed design. Lawhern *et al.* introduced EEGNet, a compact CNN that uses depthwise and separable convolutions to encode temporal filtering and spatial channel mixing efficiently [1]. EEGNet established that an EEG-specific compact architecture can generalize across several BCI paradigms, but it still learns electrode relations through convolution rather than an explicit graph.

Altaheri *et al.* proposed a physics-informed attention temporal convolutional network for motor-imagery decoding [2]. Its combination of attention and temporal convolution demonstrates the value of local and long-range temporal structure. However, channel interaction remains attention-based and does not directly encode an anatomical or functional electrode topology.

Song *et al.* developed EEG Conformer, which combines convolutional extraction of local features with transformer self-attention for global dependencies and brain-topography visualization [3]. The model supports the use of global context, yet self-attention and graph connectivity are not equivalent: attention is learned from representations, whereas a graph can impose a

reproducible structural prior.

Wang *et al.* proposed EEG_GENet, adding feature-level graph embedding to EEGNet using graphs derived from electrode positions or Pearson correlation [4]. This work directly supports graph-aware EEG modeling. Its principal limitation for the present problem is that a single static graph definition may not jointly capture stable scalp geometry and training-population functional dependence.

Ju and Guan introduced Graph-CSPNet, which performs graph learning on symmetric positive-definite manifolds to preserve covariance geometry across time-frequency representations [5]. The method demonstrates the power of geometric graph learning, but it is substantially more complex than an ordinary node-feature GNN and requires specialized covariance-manifold processing.

Together, these studies suggest that effective motor-EEG decoding should preserve temporal structure, model global dependencies, represent inter-electrode relations explicitly, and evaluate subject variability carefully. The present work differs by using a leakage-safe hybrid adjacency and by comparing its proposed ST-GNN against classical, ensemble, neural, and convolutional baselines under one fixed subject-holdout protocol.

3. Proposed Method

3.1. Problem formulation

Each accepted trial is represented by a normalized signal matrix

$$X \in \mathbb{R}^{N \times T}, \quad (1)$$

where $N = 64$ electrodes and $T = 400$ samples after resampling and cropping. The target $y \in \{0, 1\}$ denotes left-fist or right-fist execution. The electrode graph is $G = (V, E, A)$, where $|V| = 64$ and $A \in \mathbb{R}^{64 \times 64}$ is the normalized hybrid adjacency.

3.2. Graph construction

3.2.1 Nodes

Every EEG electrode is a node. This preserves the original 64-channel montage and avoids forcing the electrodes onto an artificial rectangular grid. A node contains both its raw temporal signal and ten handcrafted features.

3.2.2 Anatomical edges

The anatomical graph connects each electrode to its six nearest neighbours in the standard 10-05 montage. Let d_{ij} denote the Euclidean distance between electrode coordinates. For selected neighbours,

$$A_{ij}^{(a)} = \exp \left[-\frac{d_{ij}^2}{2\sigma_i^2} \right], \quad (2)$$

where σ_i is the median distance from node i to its selected neighbours. The matrix is symmetrized. This graph encodes stable scalp proximity and produced 221 undirected non-zero edges.

3.2.3 Functional edges

The functional graph is calculated from training subjects only. Up to 1,000 training trials are concatenated across time, and absolute Pearson correlations are computed between channels. For each node, the eight strongest correlations are retained, followed by symmetrization. This produced 327 undirected non-zero edges. Because neither validation nor test trials contribute to this calculation, the graph does not leak held-out-subject information.

3.2.4 Hybrid edge weights

The two graphs are combined as

$$A^{(h)} = 0.40A^{(a)} + 0.60A^{(f)}. \quad (3)$$

Self-loops are added and symmetric degree normalization is applied:

$$\hat{A} = D^{-1/2}(A^{(h)} + I)D^{-1/2}. \quad (4)$$

The hybrid design fits EEG because physical proximity supplies a stable spatial prior, while training-only correlation supplies data-adaptive coupling. The interpretation remains statistical: an edge does not prove a direct neural pathway because scalp EEG is affected by volume conduction and referencing.

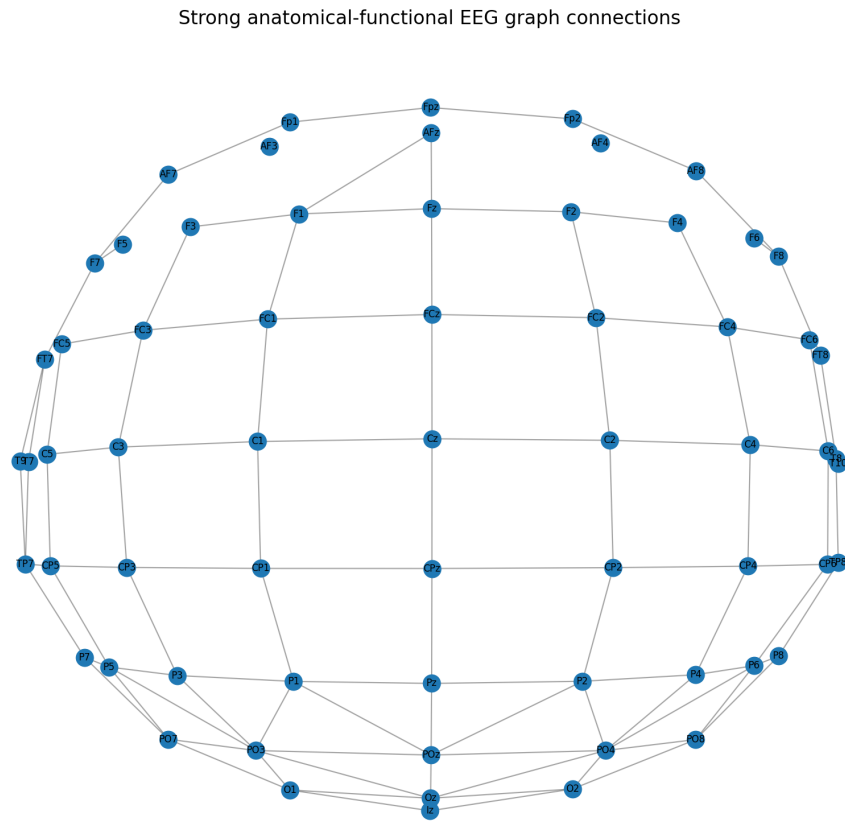


Figure 1: Strong anatomical-functional connections among the 64 EEG nodes.

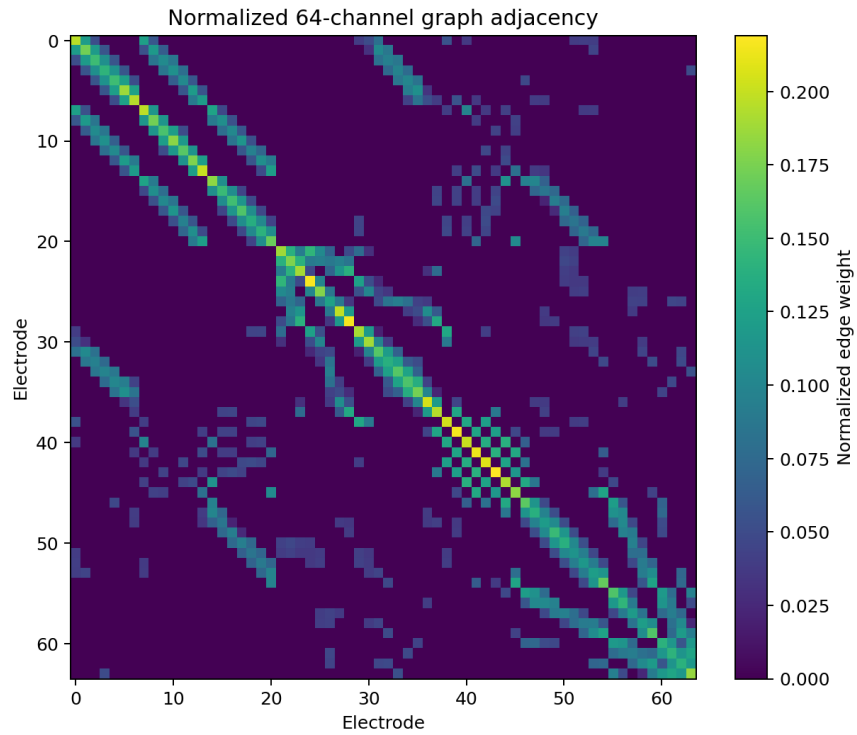


Figure 2: Normalized 64-channel hybrid adjacency used by the ST-GNN.

3.3. Node features

The feature branch uses ten features per electrode. Five are log-relative spectral powers: mu (8–12 Hz), low beta (12–16 Hz), mid beta (16–20 Hz), high beta (20–24 Hz), and upper beta (24–30 Hz). Five are time-domain descriptors: log variance, log RMS amplitude, log line length, zero-crossing rate, and Hjorth mobility. Feature normalization statistics are estimated from the training set only.

3.4. Hybrid ST-GNN architecture

Figure 3 gives the architecture implemented in the notebook. The temporal branch applies three spatio-temporal graph blocks. Each block performs temporal convolution, graph propagation with $\hat{A}$, learnable gated fusion of graph-propagated and unpropagated features, pointwise mixing, depthwise temporal refinement, residual addition, average pooling, and dropout. The block widths are 48, 80, and 112 channels with temporal kernels 15, 11, and 7.

After the third block, temporal averaging yields one 112-dimensional representation per node. Attention pooling and max pooling produce a 224-dimensional raw-signal summary. In parallel, the 10-dimensional handcrafted features are projected to 64 dimensions, propagated through the graph, and summarized by attention and max pooling to form a 128-dimensional feature summary. The two summaries are concatenated and passed through a fusion MLP of 256 and 96 hidden units before the two-class output.

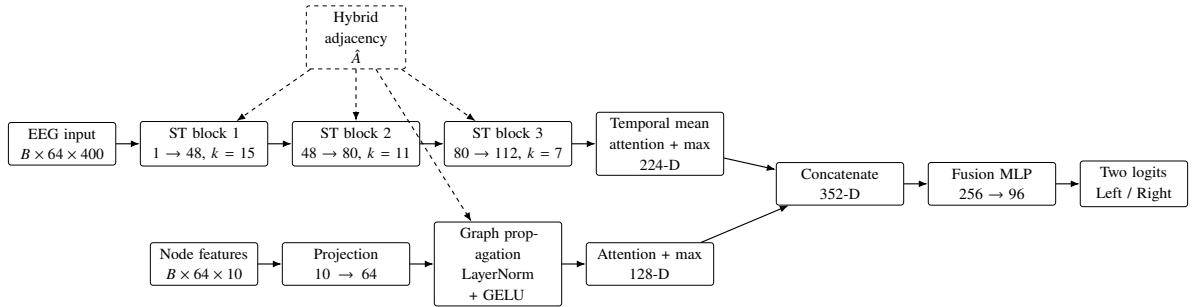


Figure 3: Implemented hybrid ST-GNN: raw temporal and handcrafted node-feature branches share the training-only hybrid adjacency before fusion.

3.5. Training settings

The ST-GNN used AdamW, an initial maximum learning rate of 2×10^{-3} , weight decay 3×10^{-4} , batch size 64, dropout 0.20, label smoothing 0.02, gradient clipping at 3.0, mixed precision, and a cosine OneCycle learning-rate schedule. Training was capped at 20 epochs with early stopping after six non-improving epochs. Class weights were calculated from training labels. Mild training-only augmentation included amplitude scaling, Gaussian noise, temporal shifts, and short temporal masking.

4. Experimental Setup

4.1. Dataset and task

The experiment used version 1.0.0 of the PhysioNet EEG Motor Movement/Imagery Dataset [6], collected with BCI2000 [7]. The source dataset contains 64-channel recordings at 160 Hz from 109 volunteers. This experiment selected physical left-versus-right fist execution from runs 3, 7, and 11. The T1 annotation was mapped to left fist and T2 to right fist for these runs.

The preprocessing audit found 1,526 canonical EDF files in the source mirror and selected 327 task-relevant files. After channel checks, epoch-boundary checks, non-finite-value checks, and artifact rejection, the final dataset contained 4,548 accepted trials from 108 subjects and 323 recordings. The class distribution was nearly balanced: 2,276 left-fist trials and 2,272 right-fist trials. A total of 379 events or files were rejected.



Figure 4: Accepted-trial class distribution for the left-versus-right execution task.

4.2. Preprocessing

The reproducible signal pipeline was:

1. normalize channel names to the standard 10-05 montage and reorder all records to the same 64-channel sequence;
2. apply common-average reference;
3. apply an 8–30 Hz IIR band-pass filter;
4. epoch each T1/T2 event from -1 to $+4$ s;
5. subtract each epoch’s channel-wise temporal mean;
6. reject non-finite epochs or epochs with maximum channel peak-to-peak amplitude above $350 \mu\text{V}$;
7. resample to 100 Hz, producing 500 samples for the full epoch;
8. crop 0–4 s for modeling, producing 400 samples per channel.

4.3. Subject-independent split

A grouped 70/15/15 split was performed with subjects as groups. No subject appeared in more than one partition.

Table 1: Subject-holdout split used in all reported comparisons.

Split	Trials	Subjects	Left fist	Right fist
Training	3,157	75	1,577	1,580
Validation	683	16	343	340
Test	708	17	356	352

4.4. Leakage safeguards

The following safeguards were applied before any test evaluation:

- subjects were split before fitting preprocessing transformations;
- the functional adjacency was estimated from training trials only;
- median imputation, standardization, and ANOVA feature selection were fitted on training features only and then applied unchanged to validation and test data;
- class oversampling was limited to the training feature matrix;
- deep-model augmentation was applied only by the training data loader;
- class weights were derived from training labels;
- checkpoints and the baseline to beat were selected using validation metrics only;
- the test set was used for final reporting, not model selection;
- measured subject overlaps were zero for train-validation, train-test, and validation-test pairs.

4.5. Baseline feature preparation

The classical models used the 64-by-10 node-feature tensor flattened to 640 features, plus 130 left-minus-right homologous asymmetry features from 13 electrode pairs, for 770 raw features. A training-fitted median imputer handled missing values, a standard scaler normalized the features, and ANOVA F-selection retained the best 384 features. Random oversampling balanced the training set for k-NN, Gradient Boosting, MLP, and XGBoost. The other classifiers used built-in class weighting where supported.

4.6. Models and settings

Table 2: Executed baselines and principal settings. LightGBM was not run in the supplied experiment.

Model	Principal settings
Logistic Regression	$C = 1.0$, balanced classes, liblinear solver, 3,000 maximum iterations.
RBF SVM	$C = 3.0$, RBF kernel, scale gamma, balanced classes, probability calibration enabled.
k-NN	9 neighbours, distance weighting, Euclidean distance.
Decision Tree	Maximum depth 12, minimum leaf size 4, balanced classes.
Random Forest	500 trees, square-root feature sampling, minimum leaf size 2, balanced subsampling.
Gradient Boosting	180 estimators, learning rate 0.04, depth 3, subsample 0.85.
Extra Trees	600 trees, 70% feature sampling, balanced classes.
MLP	Hidden layers 256 and 64, ReLU, batch size 64, learning rate 8×10^{-4} , maximum 20 iterations, early stopping.
XGBoost	350 trees, depth 5, learning rate 0.035, subsample 0.85, column sample 0.80.
Motor-area CSP-LDA	Up to 16 CSP components from FC/C/CP motor-area channels; Ledoit-Wolf covariance; shrinkage LDA.
1D-CNN	Convolution widths 96, 128, 160; kernels 15, 9, 7; grouped convolutions; global average pooling; 64-unit classifier.
Hybrid ST-GNN	Three graph-temporal blocks plus node-feature graph branch and fusion MLP; settings in Section 3.5.

4.7. Metrics and computing environment

The full evaluation set comprised accuracy, balanced accuracy, macro precision, macro recall, macro-F1, weighted F1, Matthews correlation coefficient (MCC), Cohen’s kappa, log loss, and ROC-AUC. Training time and inference time were also recorded. The run used Python 3.12.13, PyTorch 2.10.0+cu128, MNE 1.12.1, NumPy 2.0.2, and an NVIDIA Tesla T4 GPU. The classical models were trained through scikit-learn-compatible implementations, and XGBoost was used when available.

5. First Results and Discussion

5.1. Validation-based model selection

Among the classical and feature-based baselines, Logistic Regression achieved the highest validation macro-F1 of 0.6602 and was therefore declared the **baseline to beat** before training the deep models. The later 1D-CNN reached a higher validation macro-F1 of 0.6803, making it the strongest observed baseline overall. This distinction prevents post-hoc relabeling of the predeclared comparator.

Table 5: Subject-independent test performance and computational cost of the evaluated models. Higher values are better for all performance metrics except log loss, training time, and inference time.

Model	Acc. $\uparrow$	BAcc. $\uparrow$	Prec _{macro} $\uparrow$	Rec _{macro} $\uparrow$	F1 _{macro} $\uparrow$	F1 _{weighted} $\uparrow$	MCC $\uparrow$	κ $\uparrow$	Log Loss $\downarrow$	ROC-AUC $\uparrow$	Train (s) $\downarrow$	Infer. (s) $\downarrow$
1D-CNN	0.6596	0.6596	0.6596	0.6596	0.6596	0.6596	0.3193	0.3192	1.1467	0.7079	35.193	0.680
XGBoost	0.6427	0.6426	0.6426	0.6426	0.6426	0.6426	0.2853	0.2853	0.6878	0.6780	17.490	0.007
Random Forest	0.6427	0.6425	0.6429	0.6425	0.6423	0.6424	0.2854	0.2851	0.6400	0.6818	13.204	0.173
Motor-Area CSP-LDA	0.6412	0.6412	0.6412	0.6412	0.6412	0.6412	0.2824	0.2824	0.7226	0.6676	13.860	0.028
Extra Trees	0.6384	0.6382	0.6393	0.6382	0.6376	0.6377	0.2775	0.2765	0.6413	0.6794	25.791	0.218
Gradient Boosting	0.6356	0.6354	0.6361	0.6354	0.6350	0.6351	0.2715	0.2709	0.6493	0.6748	92.521	0.004
Logistic Regression	0.6186	0.6186	0.6186	0.6186	0.6186	0.6186	0.2372	0.2372	0.8539	0.6461	0.912	0.001
MLP	0.6172	0.6171	0.6174	0.6171	0.6170	0.6170	0.2345	0.2343	1.1019	0.6499	2.328	0.002
RBF-SVM	0.6116	0.6116	0.6116	0.6116	0.6116	0.6116	0.2233	0.2232	0.7210	0.6578	9.163	0.649
Decision Tree	0.5777	0.5774	0.5780	0.5774	0.5768	0.5769	0.1555	0.1549	3.8316	0.5896	1.913	0.000
k-NN	0.5720	0.5724	0.5738	0.5724	0.5701	0.5700	0.1463	0.1447	0.9338	0.5937	0.002	0.059
Hybrid ST-GNN	0.5099	0.5090	0.5100	0.5090	0.4975	0.4980	0.0190	0.0181	0.6943	0.4882	158.898	4.369

Table 3: Validation performance used for model selection.

Model	Accuracy	Balanced Acc.	Macro-F1	ROC-AUC
1D-CNN	0.6808	0.6807	0.6803	0.7338
Logistic Regression	0.6603	0.6604	0.6602	0.6970
MLP	0.6530	0.6531	0.6529	0.6989
XGBoost	0.6442	0.6443	0.6440	0.6889
Motor-area CSP-LDA	0.6428	0.6429	0.6422	0.6931
Extra Trees	0.6413	0.6413	0.6413	0.6790
Gradient Boosting	0.6369	0.6369	0.6369	0.6915
Random Forest	0.6325	0.6325	0.6325	0.6814
RBF SVM	0.6325	0.6328	0.6311	0.6869
k-NN	0.5857	0.5861	0.5815	0.6502
Decision Tree	0.5783	0.5783	0.5783	0.5814
Hybrid ST-GNN	0.5461	0.5456	0.5403	0.5291

5.2. Subject-independent test comparison

The best honest subject-independent test performance came from the 1D-CNN, with 0.6596 accuracy, 0.6596 macro-F1, and 0.7079 ROC-AUC. XGBoost and Random Forest followed at 0.6427 accuracy. Logistic Regression, the predeclared classical baseline to beat, achieved 0.6186 accuracy and macro-F1. The proposed Hybrid ST-GNN achieved 0.5099 accuracy, 0.4975 macro-F1, and 0.4882 ROC-AUC. It therefore underperformed Logistic Regression by 10.88 percentage points in accuracy and the 1D-CNN by 14.97 points.

Table 4: Primary first-results comparison.

Role	Model	Accuracy	Bal. Acc.	Macro-F1	Train s
Predeclared classical baseline	Logistic Regression	0.6186	0.6186	0.6186	0.91
Strongest observed baseline	1D-CNN	0.6596	0.6596	0.6596	35.19
Proposed model	Hybrid ST-GNN	0.5099	0.5090	0.4975	158.90

5.3. Confusion matrix and learning behaviour

The ST-GNN predicted 236 of 356 left-fist trials correctly but only 125 of 352 right-fist trials correctly. Left-fist recall was 0.6629, whereas right-fist recall was 0.3551. This asymmetric behaviour explains why overall accuracy was close to chance despite apparently reasonable left-class recall.

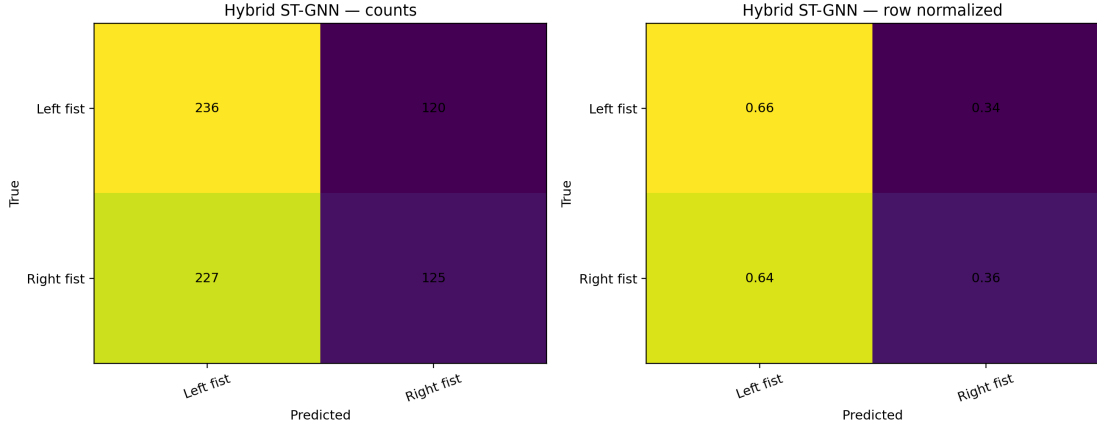
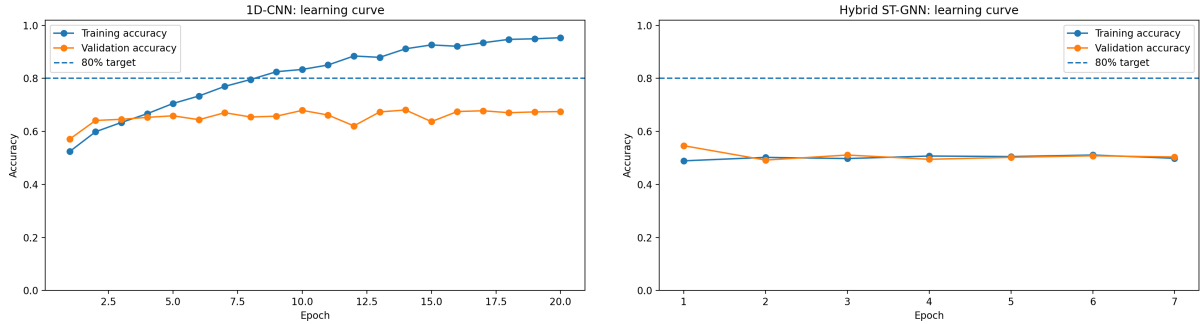


Figure 5: Hybrid ST-GNN test confusion matrices in counts and row-normalized form.

The 1D-CNN's best validation checkpoint occurred at epoch 14. Its training accuracy was approximately 0.912 and validation accuracy approximately 0.681, a gap of 0.231 that indicates overfitting. The ST-GNN's best checkpoint occurred at epoch 1, with training accuracy 0.489 and validation accuracy 0.546. Its low training and validation performance indicates underfitting rather than overfitting.



(a) 1D-CNN: improving training accuracy with a widening generalization gap. (b) Hybrid ST-GNN: training and validation remain near chance.

Figure 6: Learning curves for the strongest baseline and the proposed model.

5.4. Physiological diagnostic

The exploratory mu and beta task-to-baseline maps showed positive mean power changes for both classes: approximately 9.57–9.65 dB in mu and 8.73–8.80 dB in beta. Under the stated $10 \log_{10}(P_{\text{task}}/P_{\text{baseline}})$ definition, positive values indicate increased task power rather than conventional ERD. Therefore, the current maps should not be presented as evidence of

class-specific contralateral desynchronization without checking the baseline interval, cue-related artifacts, scaling, and topographic normalization.

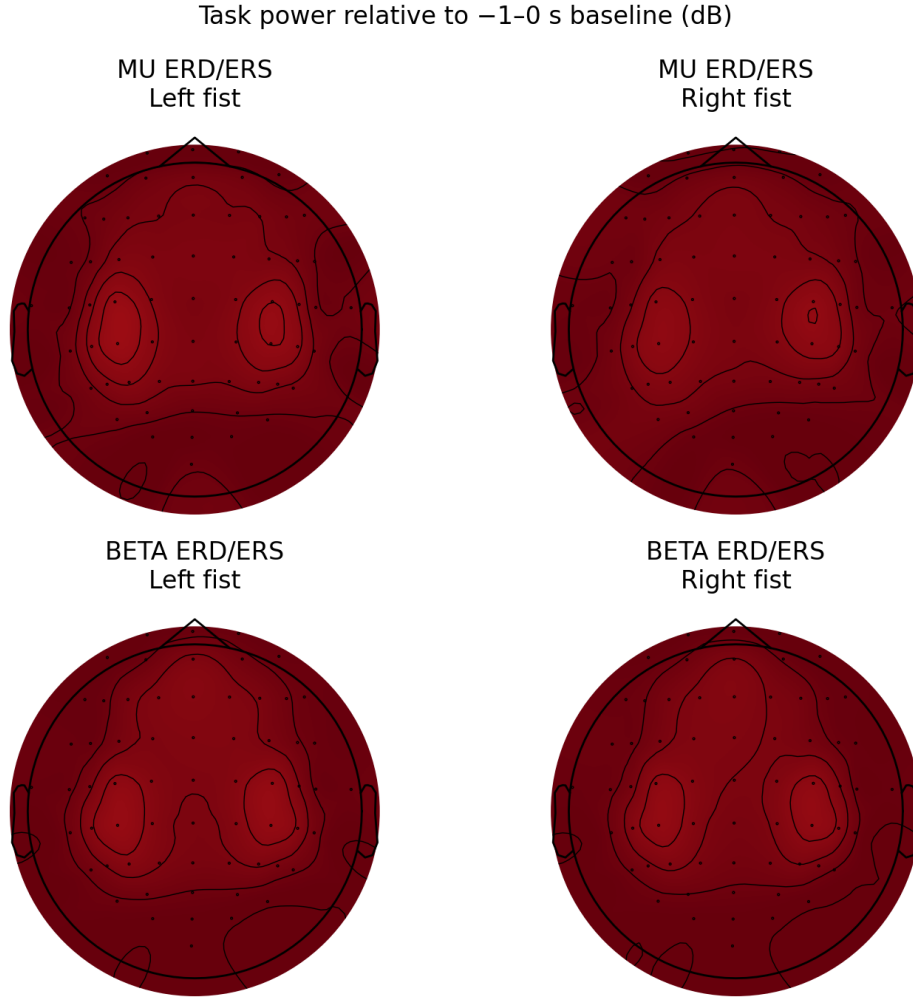


Figure 7: Exploratory task-to-baseline mu and beta power maps. Their uniformly positive values require further physiological validation.

5.5. Interpretation and next experiments

The first result does not support the claim that the proposed hybrid graph improves unseen-subject classification. Three explanations are plausible. First, the static population-level functional graph may average away subject-specific relationships. Second, repeated graph mixing and temporal pooling may oversmooth signals before discriminative lateralized patterns are learned. Third, the 20-epoch cap and regularized dual-branch model may be too restrictive for the parameter count.

The next defensible experiments are controlled ablations rather than test-set tuning: anatomical-only versus functional-only versus hybrid adjacency; identity and randomized graph controls; lower graph depth; removal of the graph gate; subject-adaptive or trial-dynamic adjacency; temporal-only and feature-only branches; reduced pooling; alternative 7–35 Hz and individualized mu/beta bands; and validation-based tuning of dropout, learning rate, and hidden width. These changes must be selected on new validation folds, followed by a single untouched test evaluation.

6. Conclusion

This report presents a reproducible leakage-safe comparison of classical models, ensembles, MLP, CSP-LDA, 1D-CNN, and a hybrid ST-GNN for left-versus-right fist execution from 64-channel EEG. The experiment used 108 subjects, subject-disjoint partitions, training-only preprocessing, training-only functional graph construction, and a complete metric set.

Logistic Regression was the predeclared classical baseline to beat, but the 1D-CNN became the strongest observed baseline with 65.96% test accuracy. The Hybrid ST-GNN reached 50.99% accuracy and showed clear underfitting. The scientifically correct conclusion is therefore not that the GNN is superior, but that the current graph and optimization design did not transfer effectively to unseen subjects. The negative result is still useful because it identifies precise ablations and safeguards for the next iteration while preserving an honest subject-independent benchmark.

References

- [1] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, “EEGNet: A compact convolutional neural network for EEG-based brain-computer interfaces,” *Journal of Neural Engineering*, vol. 15, no. 5, Art. no. 056013, 2018, doi: 10.1088/1741-2552/aace8c.
- [2] H. Altaheri, G. Muhammad, and M. Alsulaiman, “Physics-informed attention temporal convolutional network for EEG-based motor imagery classification,” *IEEE Transactions on Industrial Informatics*, vol. 19, no. 2, pp. 2249–2258, 2023, doi: 10.1109/TII.2022.3197419.
- [3] Y. Song, Q. Zheng, B. Liu, and X. Gao, “EEG Conformer: Convolutional Transformer for EEG decoding and visualization,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 31, pp. 710–719, 2023, doi: 10.1109/TNSRE.2022.3230250.
- [4] H. Wang, H. Yu, and H. Wang, “EEG_GENet: A feature-level graph embedding method for motor imagery classification based on EEG signals,” *Biocybernetics and Biomedical Engineering*, vol. 42, no. 3, pp. 1023–1040, 2022, doi: 10.1016/j.bbe.2022.08.003.
- [5] C. Ju and C. Guan, “Graph neural networks on SPD manifolds for motor imagery classification: A perspective from the time-frequency analysis,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 12, pp. 17701–17715, 2024, doi: 10.1109/TNNLS.2023.3307470.
- [6] G. Schalk, “EEG Motor Movement/Imagery Dataset,” PhysioNet, Version 1.0.0, 2009, doi: 10.13026/C28G6P.
- [7] G. Schalk, D. J. McFarland, T. Hinterberger, N. Birbaumer, and J. R. Wolpaw, “BCI2000: A general-purpose brain-computer interface system,” *IEEE Transactions on Biomedical Engineering*, vol. 51, no. 6, pp. 1034–1043, 2004, doi: 10.1109/TBME.2004.827072.